Chapter 9. Automorphisms and Galois Theory

Field automorphisms, isomorphism extension, splitting fields, separability, Galois theory, cyclotomic extensions, the quintic

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Throughout, $\overline{F}$ is a fixed algebraic closure of F and all algebraic extensions of F are taken inside it.

9.1 Automorphisms of fields (§48)

Motivation. Galois studied a polynomial through the automorphisms of a field extension fixing the base field, which permute the roots.

α, β algebraic over F are **conjugate over** F if $\operatorname{irr}(\alpha, F) = \operatorname{irr}(\beta, F)$.

Theorem 9.1 (48.3, conjugation isomorphisms)

If α, β are algebraic over F with $\deg(\alpha, F) = n$, the map $\psi_{\alpha,\beta}(c_0 + c_1\alpha + \cdots + c_{n-1}\alpha^{n-1}) = c_0 + c_1\beta + \cdots + c_{n-1}\beta^{n-1}$ is an isomorphism $F(\alpha) \to F(\beta)$ iff α and β are conjugate over F.

Proof. If $\psi_{\alpha,\beta}$ is an isomorphism, it fixes F, so $\operatorname{irr}(\alpha, F)(\beta) = \psi(\operatorname{irr}(\alpha, F)(\alpha)) = 0$, hence $\operatorname{irr}(\beta, F) = \operatorname{irr}(\alpha, F)$. Conversely, if $p = \operatorname{irr}(\alpha, F) = \operatorname{irr}(\beta, F)$, the evaluation maps give isomorphisms $F[x]/\langle p\rangle \to F(\alpha)$ and $F[x]/\langle p\rangle \to F(\beta)$ (Theorem 29.18); $\psi_{\alpha,\beta}$ is the composite of the inverse of the first with the second. ■

Corollary 9.2 (48.5)

An isomorphism of $F(\alpha)$ onto a subfield of $\overline{F}$ fixing F maps α to a conjugate of α; conversely, for each conjugate β there is exactly one such isomorphism with $\alpha \mapsto \beta$, namely $\psi_{\alpha,\beta}$.

Proof. $\psi(\alpha)$ is a zero of $\operatorname{irr}(\alpha, F)$; and an isomorphism fixing F is determined by the image of α. ■

Corollary 9.3 (48.6)

If $f \in \mathbb{R}[x]$ and $f(a + bi) = 0$, then $f(a - bi) = 0$.

Proof. Complex conjugation is $\psi_{i,-i} : \mathbb{R}(i) \to \mathbb{R}(i)$, fixing the coefficients of f. ■

An isomorphism of E onto itself is an **automorphism**. For automorphisms σ_i, the **fixed field** is $E_{\{\sigma_i\}} = \{a : \sigma_i(a) = a\ \forall i\}$.

Theorem 9.4 (48.11)

$E_{\{\sigma_i\}}$ is a subfield of E.

Proof. Each σ_i preserves sums, products, negatives and inverses. ■

Theorem 9.5 (48.14)

The automorphisms of a field form a group under composition.

Theorem 9.6 (48.15)

For $F \leq E$, the automorphisms of E fixing F form a subgroup $G(E/F)$, whose fixed field contains F.

Proof. Composites and inverses of automorphisms fixing F fix F. ■

Theorem 9.7 (48.19, Frobenius)

For a finite field F of characteristic p, $\sigma_p(a) = a^p$ is an automorphism of F with fixed field $\mathbb{Z}_p$.

Proof. σ_p is a ring homomorphism (Lemma 33.9), one-to-one ($a^p = 0$ only for $a = 0$), hence onto the finite set F. The fixed elements are the zeros of $x^p - x$, i.e. the p elements of $\mathbb{Z}_p$ (Fermat). ■

9.2 The isomorphism extension theorem (§49)

Motivation. The isomorphism extension theorem extends isomorphisms to larger fields, and the number of extensions $\{E : F\}$ is the key invariant.

Theorem 9.8 (49.3, isomorphism extension theorem)

Let E be algebraic over F, $\sigma : F \to F'$ an isomorphism, and $\overline{F'}$ an algebraic closure of F'. Then σ extends to an isomorphism τ of E onto a subfield of $\overline{F'}$.

Proof. Consider pairs (L, λ) with $F \leq L \leq E$ and $\lambda : L \to \overline{F'}$ an isomorphism onto its image extending σ, ordered by extension. The union of a chain is again such a pair, so Zorn's lemma gives a maximal (K, λ). If $K \neq E$, take $\alpha \in E - K$ and $p = \operatorname{irr}(\alpha, K)$; the polynomial $\lambda(p)$ is irreducible over $\lambda[K]$ and has a zero $\beta \in \overline{F'}$. As in Theorem 9.1, $K(\alpha) \cong K[x]/\langle p \rangle \cong \lambda[K][x]/\langle \lambda(p) \rangle \cong \lambda[K](\beta)$ gives an extension of λ to $K(\alpha)$, contradicting maximality. So $K = E$. ■

Corollary 9.9 (49.4)

If $E \leq \overline{F}$ is algebraic over F and $\alpha, \beta \in E$ are conjugate over F, then $\psi_{\alpha,\beta}$ extends to an isomorphism of E onto a subfield of $\overline{F}$.

Corollary 9.10 (49.5)

Any two algebraic closures of F are isomorphic by an isomorphism fixing F.

Proof. Extend id_F to $\tau : \overline{F} \to \overline{F}'$. $\tau[\overline{F}]$ is algebraically closed, and $\overline{F}'$ is algebraic over it, so $\overline{F}' = \tau[\overline{F}]$ (Corollary 31.16). ■

The same argument shows: every isomorphism of $\overline{F}$ into itself fixing F is an automorphism; and every isomorphism $\sigma : F \to F'$ extends to an isomorphism $\overline{F} \to \overline{F'}$.

Theorem 9.11 (49.7)

Let E be a finite extension of F, $\sigma : F \to F'$ an isomorphism and $\overline{F'}$ an algebraic closure of F'. The number of extensions of σ to an isomorphism of E onto a subfield of $\overline{F'}$ is finite, independent of σ, F' and $\overline{F'}$, and at most $[E : F]$.

Proof. First let $E = F(\alpha)$, $p = \operatorname{irr}(\alpha, F)$. Extensions of σ correspond bijectively to the zeros of $\sigma(p)$ in $\overline{F'}$ (an extension sends α to such a zero; conversely each zero β gives one, as in the proof of Theorem 9.8). Extending σ to an isomorphism $\overline{F} \to \overline{F'}$ (remark after Corollary 9.10) carries the zeros of p bijectively to those of $\sigma(p)$, so their number m equals the number of distinct zeros of p in $\overline{F}$; $m \leq \deg p = [F(\alpha) : F]$. In general, write $E = F(\alpha_1, \ldots, \alpha_n)$; each extension of σ to

$F(\alpha_1,\dots,\alpha_{k-1})$ has the same number m_k of extensions to $F(\alpha_1,\dots,\alpha_k)$, by the simple case; so the total is $m_1\cdots m_n \le \prod[F(\alpha_1,\dots,\alpha_k):F(\alpha_1,\dots,\alpha_{k-1})] = [E:F]$. ■

The number is the **index** $\{E:F\}$.

Corollary 9.12 (49.10)

If $F \le E \le K$ with K finite over F, then $\{K:F\} = \{K:E\}\{E:F\}$.

Proof. Each of the $\{E:F\}$ extensions of id_F to E has exactly $\{K:E\}$ extensions to K (independence in Theorem 9.11). ■

9.3 Splitting fields (§50)

Motivation. Splitting fields contain all conjugates of their elements ($\mathbb{Q}(\sqrt[3]{2})$ does not); for them every embedding into $\overline{F}$ is an automorphism.

E with $F \le E \le \overline{F}$ is a **splitting field over** F if it is the smallest subfield of $\overline{F}$ containing F and all zeros in $\overline{F}$ of each polynomial of some set $\{f_i\} \subseteq F[x]$.

Theorem 9.13 (50.3)

E ($F \le E \le \overline{F}$) is a splitting field over F iff every automorphism of $\overline{F}$ fixing F maps E onto itself.

Proof. If E is generated by the zeros of the f_i, an automorphism τ fixing F permutes the zeros of each f_i, so $\tau[E] \subseteq E$, and likewise $\tau^{-1}[E] \subseteq E$. Conversely, suppose the condition holds; we show E is the splitting field of $\{\mathrm{irr}(\alpha,F) : \alpha \in E\}$, i.e. every conjugate β of every $\alpha \in E$ lies in E. $\psi_{\alpha,\beta}$ extends to an isomorphism of $\overline{F}$ into itself fixing F (Theorem 9.8), which is an automorphism (remark after Corollary 9.10); it maps E onto E, so $\beta = \tau(\alpha) \in E$. ■

Corollary 9.14 (50.6)

If $E \le \overline{F}$ is a splitting field over F, every irreducible polynomial in $F[x]$ with a zero in E splits in E.

Proof. Its zeros are the conjugates of the given zero, which lie in E by the proof of Theorem 9.13. ■

Corollary 9.15 (50.7)

If $E \le \overline{F}$ is a splitting field over F, every isomorphism of E onto a subfield of $\overline{F}$ fixing F is an automorphism of E. If E is a finite splitting field, $\{E:F\} = |G(E/F)|$.

Proof. Such an isomorphism extends to an automorphism of $\overline{F}$ (as above), which maps E onto E. ■

Example 9.1

$\mathbb{Q}(\sqrt[3]{2},\omega)$ ($\omega = e^{2\pi i/3}$) is the splitting field of x^3-2, of degree 6; $\mathbb{Q}(\sqrt[3]{2})$ is not a splitting field over $\mathbb{Q}$. The splitting field of $x^{p^n}-x$ over $\mathbb{Z}_p$ is $\mathrm{GF}(p^n)$.

9.4 Separable extensions (§51)

Motivation. Separability (no repeated roots of irreducible polynomials) makes the number of automorphisms equal the degree and yields $E = F(\alpha)$.

α is a zero of f of **multiplicity** ν if $(x-\alpha)^\nu$ divides f in $\overline{F}[x]$ but $(x-\alpha)^{\nu+1}$ does not.

Theorem 9.16 (51.2)

All zeros in $\overline{F}$ of an irreducible $f \in F[x]$ have the same multiplicity.

Proof. For zeros α, β, $\psi_{\alpha,\beta}$ extends to an automorphism τ of $\overline{F}$ fixing F. If $f = (x-\alpha)^\nu g$ with $g(\alpha) \neq 0$, applying τ to coefficients, $f = (x-\beta)^\nu \tau(g)$ with $\tau(g)(\beta) = \tau(g(\alpha)) \neq 0$; by unique factorization in $\overline{F}[x]$ the multiplicity of β is ν. ■

Corollary 9.17 (51.3)

An irreducible $f \in F[x]$ factors in $\overline{F}[x]$ as $a\prod_i (x-\alpha_i)^\nu$ with the α_i its distinct zeros.

Theorem 9.18 (51.6)

If E is finite over F, then $\{E:F\}$ divides $[E:F]$.

Proof. For $E = F(\alpha)$: $[F(\alpha):F] = \deg \operatorname{irr}(\alpha, F) = \nu \cdot m$ with $m = \{F(\alpha):F\}$ the number of distinct zeros (Corollary 9.17, Theorem 9.11). In general use a chain of simple extensions, Theorem 31.4 and Corollary 9.12. ■

E is a **separable extension** of F if $\{E:F\} = [E:F]$; α is **separable** over F if $F(\alpha)$ is, i.e. if $\operatorname{irr}(\alpha, F)$ has distinct zeros.

Theorem 9.19 (51.9)

For finite $F \leq E \leq K$: K is separable over F iff K is separable over E and E is separable over F.

Proof. $\{K:F\} = \{K:E\}\{E:F\}$ and $[K:F] = [K:E][E:F]$ with $\{K:E\} \leq [K:E]$ and $\{E:F\} \leq [E:F]$; equality of the products holds iff both equalities hold. ■

Corollary 9.20 (51.10)

A finite extension E of F is separable iff every $\alpha \in E$ is separable over F.

Proof. If E is separable, apply Theorem 9.19 to $F \leq F(\alpha) \leq E$. Conversely let $E = F(\alpha_1, \ldots, \alpha_n)$; $\operatorname{irr}(\alpha_i, F(\alpha_1, \ldots, \alpha_{i-1}))$ divides $\operatorname{irr}(\alpha_i, F)$, so has distinct zeros, and each step is separable; apply Theorem 9.19. ■

Lemma 9.21 (51.11)

If $f = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \overline{F}[x]$, $f^m \in F[x]$ and $m \cdot 1 \neq 0$ in F, then $f \in F[x]$.

Proof. Suppose $a_{n-1}, \ldots, a_{n-k+1} \in F$. The coefficient of x^{nm-k} in f^m is ma_{n-k} plus a polynomial expression in $a_{n-1}, \ldots, a_{n-k+1}$ with integer coefficients; it lies in F, so $ma_{n-k} \in F$ and $a_{n-k} \in F$. Induct on k. ■

A field is **perfect** if every finite extension is separable.

Theorem 9.22 (51.13)

Every field of characteristic 0 is perfect.

Proof. Let f be monic irreducible over F with distinct zeros α_i of multiplicity ν, and $g = \prod(x-\alpha_i)$. Then $g^\nu = f \in F[x]$ and $\nu \cdot 1 \neq 0$, so $g \in F[x]$ (Lemma 9.21); $g \mid f$ and f irreducible force $\nu = 1$. So every algebraic element is separable; apply Corollary 9.20. ■

Theorem 9.23 (51.14)

Every finite field is perfect.

Proof. If F is finite and α is algebraic over F, then $F(\alpha)$ is finite, of p^n elements, so α is a zero of $x^{p^n} - x$, which has distinct zeros (Lemma 33.8); $\operatorname{irr}(\alpha, F)$ divides it, so has distinct zeros. ■

Theorem 9.24 (51.15, primitive element theorem)

A finite separable extension E of F is simple: $E = F(\gamma)$.

Proof. If F is finite, use Corollary 33.6. Let F be infinite; by induction on the number of generators it suffices to treat $E = F(\alpha, \beta)$. Let $\sigma_1, \ldots, \sigma_n$ be the $n = \{E : F\} = [E : F]$ distinct isomorphisms of E into $\overline{F}$ fixing F. For $i \neq j$, $(\sigma_i\alpha, \sigma_i\beta) \neq (\sigma_j\alpha, \sigma_j\beta)$, so $\sigma_i\alpha + c\sigma_i\beta = \sigma_j\alpha + c\sigma_j\beta$ for at most one $c \in F$. Choose c outside these finitely many values and put $\gamma = \alpha + c\beta$. The σ_i restricted to $F(\gamma)$ are n distinct isomorphisms (they differ on γ), so $[F(\gamma) : F] \geq \{F(\gamma) : F\} \geq n = [E : F]$, and $F(\gamma) = E$. ■

Corollary 9.25 (51.16)

A finite extension of a field of characteristic 0 is simple.

9.5 Totally inseparable extensions (§52)

Motivation. In characteristic p separability can fail ($x^p - t$ over $\mathbb{Z}_p(t)$); every algebraic extension is separable followed by totally inseparable.

A finite extension E of F is **totally inseparable** if $\{E : F\} = 1 < [E : F]$; $\alpha \notin F$ is totally inseparable if $F(\alpha)$ is.

Theorem 9.26 (52.3)

For finite $F < E < K$: K is totally inseparable over F iff K is over E and E is over F.

Proof. $\{K : F\} = \{K : E\}\{E : F\}$ is 1 iff both factors are 1. ■

Corollary 9.27 (52.4)

A finite extension $E > F$ is totally inseparable iff every $\alpha \in E - F$ is totally inseparable over F.

Proof. If E is totally inseparable, then $\{E:F\}=\{E:F(\alpha)\}\{F(\alpha):F\}=1$ forces $\{F(\alpha):F\}=1$, while $[F(\alpha):F]>1$. Conversely if $E=F(\alpha_1,\dots,\alpha_n)$ with each $\operatorname{irr}(\alpha_i,F)$ having one distinct zero, then so does $\operatorname{irr}(\alpha_i,F(\alpha_1,\dots,\alpha_{i-1}))$, which divides it; so every step has index 1, and $\{E:F\}=1$. ■

Theorem 9.28 (52.5)

Let F have characteristic $p\neq 0$ and E be finite over F. Then $\alpha\in E-F$ is totally inseparable over F iff $\alpha^{p^t}\in F$ for some $t\geq 1$. Moreover the separable elements of E form a subfield K (the **separable closure** of F in E), and E is totally inseparable over K unless $E=K$.

Proof. If $\alpha^{p^t}=a\in F$, then $x^{p^t}-a=(x-\alpha)^{p^t}$ has the single zero α, so $\operatorname{irr}(\alpha,F)$ does too. Conversely, if $\operatorname{irr}(\alpha,F)=(x-\alpha)^m$, write $m=p^tk$ with $p\nmid k$; then $(x-\alpha)^m=(x^{p^t}-\alpha^{p^t})^k$, and Lemma 9.21 gives $x^{p^t}-\alpha^{p^t}\in F[x]$; $t\geq 1$ since otherwise $k=m$ and $x-\alpha\in F[x]$.

Separable elements form a subfield by Corollary 9.20 applied to $F(\alpha,\beta)$. For $\alpha\in E$, $f=\operatorname{irr}(\alpha,F)$ can be written $g(x^{p^t})$ with g irreducible and $g'\neq 0$ (if $f'=0$ then $f\in F[x^p]$; iterate), and then g has distinct zeros ($\gcd(g,g')=1$), so α^{p^t} is separable, i.e. lies in K. By the first part (over K), E is totally inseparable over K. ■

9.6 Galois theory (§53)

Motivation. For a finite normal extension, intermediate fields correspond to subgroups of the Galois group, reversing inclusions.

A finite extension K of F is a **finite normal extension** if it is a separable splitting field over F; then $G(K/F)$ is its **Galois group**, and $|G(K/F)|=\{K:F\}=[K:F]$ (Corollary 9.15).

Theorem 9.29 (53.2)

If K is finite normal over F and $F\leq E\leq K$, then K is finite normal over E, and $G(K/E)$ is the subgroup of $G(K/F)$ of automorphisms fixing E. Two automorphisms $\sigma,\tau\in G(K/F)$ induce the same isomorphism of E into $\overline{F}$ iff they lie in the same left coset of $G(K/E)$.

Proof. K is separable over E (Theorem 9.19), and the splitting field over F of a set of polynomials is also their splitting field over E. $\sigma|_E=\tau|_E$ iff $\sigma^{-1}\tau$ fixes E. ■

Theorem 9.30 (53.6, main theorem of Galois theory)

Let K be finite normal over F, $G=G(K/F)$, and for $F\leq E\leq K$ let $\lambda(E)=G(K/E)$. Then:
(1) λ is a bijection from the intermediate fields onto the subgroups of G;
(2) $\lambda(E)=G(K/E)$ and $E=K_{G(K/E)}$;
(3) for $H\leq G$, $\lambda(K_H)=H$;
(4) $[K:E]=|\lambda(E)|$ and $[E:F]=(G:\lambda(E))$;
(5) E is normal over F iff $\lambda(E)\trianglelefteq G$, and then $G(E/F)\cong G/\lambda(E)$;
(6) λ reverses inclusions.

Proof. (2) Clearly $E\subseteq K_{G(K/E)}$. If $\alpha\in K-E$, $\operatorname{irr}(\alpha,E)$ has degree >1 and distinct zeros, all in K (Corollary 9.14); for a zero $\beta\neq\alpha$, $\psi_{\alpha,\beta}$ extends to an automorphism of K fixing E (Theorem 9.8 and Corollary 9.15), which moves α. So $E=K_{G(K/E)}$, and λ is one-to-one.

(3) Let $E = K_H$; then $H \leq \lambda(E)$. K is separable over E, so $K = E(\alpha)$ (Theorem 9.24). The polynomial $\prod_{\sigma\in H}(x - \sigma\alpha)$ has coefficients fixed by H, i.e. in E, and has α as a zero, so $[K : E] = \deg(\alpha, E) \leq |H|$. But $|\lambda(E)| = |G(K/E)| = [K : E]$. Hence $H = \lambda(E)$; this proves surjectivity in (1).

(4) $|\lambda(E)| = [K : E]$, and $[E : F] = [K : F]/[K : E] = |G|/|\lambda(E)|$.

(5) For $\sigma \in G$, $\lambda(\sigma[E]) = \sigma\lambda(E)\sigma^{-1}$. E is a splitting field over F iff every isomorphism of E into $\overline{F}$ fixing F maps E onto E (Theorem 9.13); these isomorphisms are restrictions of elements of G (extend to K, which is a splitting field); so E is normal iff $\sigma[E] = E$ for all $\sigma \in G$, iff $\sigma\lambda(E)\sigma^{-1} = \lambda(E)$ for all σ (by (1)). Then restriction $G \to G(E/F)$ is a homomorphism, onto (every automorphism of E over F extends to K), with kernel $\lambda(E)$.

(6) $E_1 \subseteq E_2$ implies $G(K/E_2) \leq G(K/E_1)$, and conversely by (2). ■

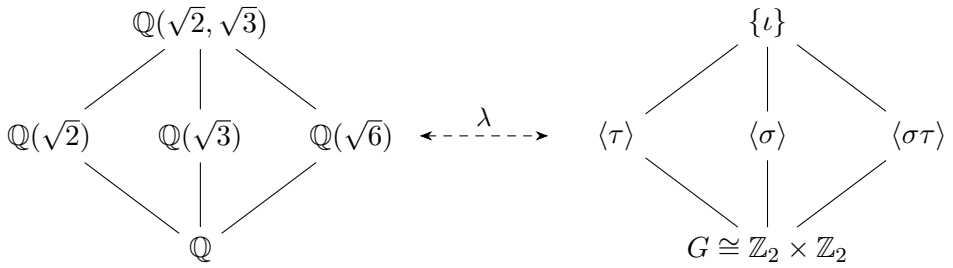


Figure. *The Galois correspondence for $K = \mathbb{Q}(\sqrt{2},\sqrt{3})$, with $\sigma : \sqrt{2} \mapsto -\sqrt{2}$ and $\tau : \sqrt{3} \mapsto -\sqrt{3}$. Each intermediate field is the fixed field of the subgroup in the same position; the lattices are inverted (larger fields correspond to smaller groups).*

Theorem 9.31 (53.7)

If K is an extension of degree n of a finite field F of p^r elements, then $G(K/F)$ is cyclic of order n, generated by $\sigma_{p^r}(\alpha) = \alpha^{p^r}$.

Proof. K is the splitting field of $x^{p^{rn}} - x$ over F, separable (Theorem 9.23), so normal with $|G(K/F)| = n$. σ_{p^r} fixes F (whose elements satisfy $a^{p^r} = a$). If $\sigma_{p^r}^m = \mathrm{id}$ with $0 < m < n$, every $\alpha \in K$ would be a zero of $x^{p^{rm}} - x$, which has fewer than p^{rn} zeros. So σ_{p^r} has order n. ■

Example 9.2

For $K = \mathbb{Q}(\sqrt{2},\sqrt{3})$, $G(K/\mathbb{Q}) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, and the three subgroups of order 2 correspond to $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{6})$. For $K = \mathbb{Q}(\sqrt[3]{2},\omega)$, $G \cong S_3$; the normal subgroup A_3 corresponds to the normal extension $\mathbb{Q}(\omega)$.

Algebraic proof of the fundamental theorem of algebra. Let K be a finite normal extension of $\mathbb{R}$ containing i, with $G = G(K/\mathbb{R})$ of order $2^k m$, m odd. The fixed field of a Sylow 2-subgroup has odd degree m over $\mathbb{R}$; by the intermediate value theorem every real polynomial of odd degree has a real zero, so $m = 1$. Then $G(K/\mathbb{C})$ is a 2-group; if nontrivial it has a subgroup of index 2, giving a quadratic extension of $\mathbb{C}$, impossible since every complex number has a square root. So $K = \mathbb{C}$.

9.7 Illustrations of Galois theory (§54)

Motivation. Examples: Galois groups of splitting fields and their subgroup and subfield lattices side by side.

For indeterminates $y_1, \dots, y_n$ the **elementary symmetric functions** s_k are defined by $\prod(x - y_i) = x^n - s_1 x^{n-1} + s_2 x^{n-2} - \cdots + (-1)^n s_n$.

Theorem 9.32 (54.2)

Every symmetric function of $y_1, \dots, y_n$ over F is a rational function of $s_1, \dots, s_n$: $F(y_1, \dots, y_n)^{S_n} = F(s_1, \dots, s_n)$.

Proof. Let $K = F(y_1, \dots, y_n)$ and $E = F(s_1, \dots, s_n) \leq K$. K is the splitting field over E of $\prod(x - y_i) \in E[x]$, which has distinct zeros, so K is finite normal over E with $[K : E] \leq n!$ (adjoin the zeros one at a time). S_n acts on K by permuting the y_i, by automorphisms fixing E; so $|G(K/E)| \geq n!$, hence $= n!$, and $G(K/E) = S_n$. By Theorem 9.30(2), $E = K_{S_n}$, the field of symmetric functions. ■

Example 9.3

$y_1^2 + \cdots + y_n^2 = s_1^2 - 2s_2$. The Galois group of $x^4 - 2$ over $\mathbb{Q}$ is the dihedral group D_4 of order 8, and its lattice of 10 subgroups matches the lattice of intermediate fields of $\mathbb{Q}(\sqrt[4]{2}, i)$.

9.8 Cyclotomic extensions (§55)

Motivation. Adjoining roots of unity gives abelian Galois groups and Gauss's constructible regular polygons.

The splitting field of $x^n - 1$ over F is the **n-th cyclotomic extension**. Over $\mathbb{Q}$ it is $\mathbb{Q}(\zeta)$ with $\zeta = e^{2\pi i/n}$; the primitive n-th roots of unity are the ζ^k with $\gcd(k, n) = 1$, zeros of the **cyclotomic polynomial** $\Phi_n(x) = \prod_{\gcd(k,n)=1}(x - \zeta^k)$, which lies in $\mathbb{Z}[x]$ (from $x^n - 1 = \prod_{d|n} \Phi_d$ by induction).

Theorem 9.33 (55.4)

The Galois group of the n-th cyclotomic extension of $\mathbb{Q}$ has $\phi(n)$ elements and is isomorphic to the group G_n of units of $\mathbb{Z}_n$.

Proof.

Step 1. [Supplement] (Φ_n is irreducible over $\mathbb{Q}$.) Let $f = \mathrm{irr}(\zeta, \mathbb{Q})$; by Gauss's lemma $f \in \mathbb{Z}[x]$ and $x^n - 1 = fg$ with $g \in \mathbb{Z}[x]$ monic. Let p be a prime not dividing n; we show $f(\zeta^p) = 0$. Otherwise $g(\zeta^p) = 0$, so ζ is a zero of $g(x^p)$ and $f(x) \mid g(x^p)$ in $\mathbb{Z}[x]$. Reduce modulo p: $\bar{g}(x^p) = \bar{g}(x)^p$ in $\mathbb{Z}_p[x]$, so $\bar{f} \mid \bar{g}^p$ and $\bar{f}, \bar{g}$ have a common zero in $\overline{\mathbb{Z}_p}$. Then $x^n - \bar{1} = \bar{f}\bar{g}$ has a repeated zero; but its derivative nx^{n-1} is nonzero at every zero ($p \nmid n$), a contradiction. Applying this repeatedly to the prime factors of k, every ζ^k with $\gcd(k, n) = 1$ is a zero of f; hence $f = \Phi_n$ and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \phi(n)$.

Step 2. An automorphism σ is determined by $\sigma(\zeta)$, a primitive root ζ^k with $k \in G_n$; $\sigma \mapsto k$ is a homomorphism ($\sigma\tau(\zeta) = \sigma(\zeta^l) = \zeta^{kl}$), one-to-one, and onto since both groups have $\phi(n)$ elements. ■

Corollary 9.34 (55.6)

For a prime p, the Galois group of the p-th cyclotomic extension of $\mathbb{Q}$ is cyclic of order $p - 1$.

Proof. $G_p = \mathbb{Z}_p^*$ is cyclic (Corollary 23.6). ■

Theorem 9.35 (55.8, Gauss)

The regular n-gon is constructible with straightedge and compass iff all odd primes dividing n are Fermat primes $(2^{2^k}+1)$ whose squares do not divide n.

Proof. The n-gon is constructible iff $\cos(2\pi/n)$ is constructible. Since ζ is a zero of $x^2-2\cos(2\pi/n)x+1$, $[\mathbb{Q}(\zeta):\mathbb{Q}(\cos 2\pi/n)]=2$ for $n\geq 3$. If $\cos(2\pi/n)$ is constructible, its degree is a power of 2 (Corollary 32.8), so $\phi(n)$ is a power of 2. Conversely, if $\phi(n)=2^m$, the Galois group G_n is an abelian 2-group; the subgroup fixing $\mathbb{Q}(\cos 2\pi/n)$ has order 2, and the quotient, an abelian group of order 2^{m-1}, has a chain of subgroups each of index 2 (Theorem 36.8). By the Galois correspondence this gives a tower $\mathbb{Q}=L_0<L_1<\cdots<L_{m-1}=\mathbb{Q}(\cos 2\pi/n)$ of real fields with $[L_{j+1}:L_j]=2$; each $L_{j+1}=L_j(\sqrt{d_j})$ with $0<d_j\in L_j$, so $\cos(2\pi/n)$ is constructible (Theorem 32.6). Finally $\phi(n)=\prod p^{a-1}(p-1)$ is a power of 2 iff every odd $p\mid n$ has $a=1$ and $p-1$ a power of 2; and 2^s+1 prime forces s to be a power of 2. ■

So the regular 17-gon is constructible, and the regular 7-gon and 9-gon are not.

9.9 Insolvability of the quintic (§56)

Motivation. A polynomial is solvable by radicals iff its Galois group is solvable; since A_5 is simple, some quintics over $\mathbb{Q}$ are not.

An extension K of F is an **extension by radicals** if $K=F(\alpha_1,\ldots,\alpha_r)$ with $\alpha_i^{n_i}\in F(\alpha_1,\ldots,\alpha_{i-1})$; $f\in F[x]$ is **solvable by radicals** if its splitting field lies in such a K. Fields have characteristic 0 in this section.

Lemma 9.36 (56.3)

If K is the splitting field of x^n-a over F, then $G(K/F)$ is solvable.

Proof. If $a=0$ there is nothing to show. The n zeros of x^n-a are distinct (the derivative nx^{n-1} does not vanish at them); if α is one, the quotients α'/α are n distinct n-th roots of unity in K, forming a cyclic group (Corollary 23.6) with a generator ζ, a primitive n-th root of unity. So $K=F(\zeta,\alpha)$. $F(\zeta)$ is normal over F with $G(F(\zeta)/F)$ isomorphic to a subgroup of G_n, abelian. An element of $G(K/F(\zeta))$ is determined by $\alpha\mapsto\zeta^k\alpha$, and $\sigma\mapsto k$ is a one-to-one homomorphism into $\mathbb{Z}_n$; so $G(K/F(\zeta))$ is abelian. The series $\{e\}\leq G(K/F(\zeta))\trianglelefteq G(K/F)$ has abelian factors (Theorem 9.30(5)). ■

Theorem 9.37 (56.4)

If $F\leq E\leq K$ with E normal over F and K an extension of F by radicals, then $G(E/F)$ is solvable.

Proof. Let $K=F(\alpha_1,\ldots,\alpha_r)$ as above, $n=\prod n_i$, and L the splitting field over F of $(x^n-1)\prod_i \operatorname{irr}(\alpha_i,F)$; $L\supseteq K\supseteq E$ and L is normal over F. L is generated over $F(\zeta)$ (ζ a primitive n-th root of unity) by the images $\sigma(\alpha_i)$, $\sigma\in G(L/F)$, and $\sigma(\alpha_i)^{n_i}\in F(\sigma(\alpha_1),\ldots,\sigma(\alpha_{i-1}))$. Ordering these generators accordingly gives a tower $F\leq F(\zeta)=L_0\leq L_1\leq\cdots\leq L_s=L$ with $L_j=L_{j-1}(\beta_j)$, $\beta_j^{m_j}\in L_{j-1}$, $m_j\mid n$. Since L_{j-1} contains the m_j-th roots of unity, L_j is the splitting field of $x^{m_j}-\beta_j^{m_j}$ over L_{j-1}, normal with abelian Galois group (proof of Lemma 9.36). Then $G(L/L_s)\leq G(L/L_{s-1})\leq\cdots\leq G(L/L_0)\leq G(L/F)$ is a subnormal series (Theorem 9.30(5)) whose factors $G(L_j/L_{j-1})$ and $G(F(\zeta)/F)$ are abelian. So $G(L/F)$ is solvable, and so is its factor group $G(E/F)\cong G(L/F)/G(L/E)$ (factor groups of solvable groups are solvable, by Theorem 34.5 applied to a series). ■

Theorem 9.38 (56.6)

If $y_1, \dots, y_5$ are independent transcendentals over $\mathbb{Q}$, then $f(x) = \prod_{i=1}^{5}(x - y_i)$ is not solvable by radicals over $F = \mathbb{Q}(s_1, \dots, s_5)$.

Proof. By the proof of Theorem 9.32, the splitting field $\mathbb{Q}(y_1, \dots, y_5)$ has Galois group S_5 over F. S_5 is not solvable: its composition series $\{e\} < A_5 < S_5$ has the nonabelian simple factor A_5 (Theorem 15.15, Jordan–Hölder). By Theorem 9.37, f is not solvable by radicals. ■

Consequently there is no general formula by radicals for the zeros of a polynomial of degree 5; concretely, $x^5 - 4x + 2$ (irreducible by Eisenstein, with exactly three real zeros, so its Galois group contains a 5-cycle and a transposition and is S_5) is not solvable by radicals over $\mathbb{Q}$.